Farmingdale State College
EET 223T Digital Electronics
Test #1

Date: Oct 5th, 09 | Max Marks: 30
Time: 1hr | Name:____________________

1. A) Use Boolean Algebra to simplify

$$z = (B+\overline{C})(\overline{B}+C) + \overline{\overline{A}+B+\overline{C}}$$

$$= B\overline{B} + \overline{B}\,\overline{C} + BC + C\overline{C} + \overline{\overline{A}}\cdot\overline{B}\,\overline{\overline{C}}$$

$$= 0 + \overline{B}\,\overline{C} + BC + 0 + A\,\overline{B}C$$

$$= \overline{B}(\overline{C} + CA) + BC$$

$$= \overline{B}(\overline{C} + A) + BC = \overline{B}\,\overline{C} + A\overline{B} + BC$$

B) Use DeMorgan's theorem to simpilify $(M+\overline{N})(\overline{M}+N)$ (2 Marks + 1 Mark)

$$= \overline{M+\overline{N}} + \overline{\overline{M}+N}$$

$$= \overline{M}\cdot\overline{\overline{N}} + \overline{\overline{M}}\cdot\overline{N}$$

$$= \overline{M}\cdot N + M\cdot\overline{N}$$

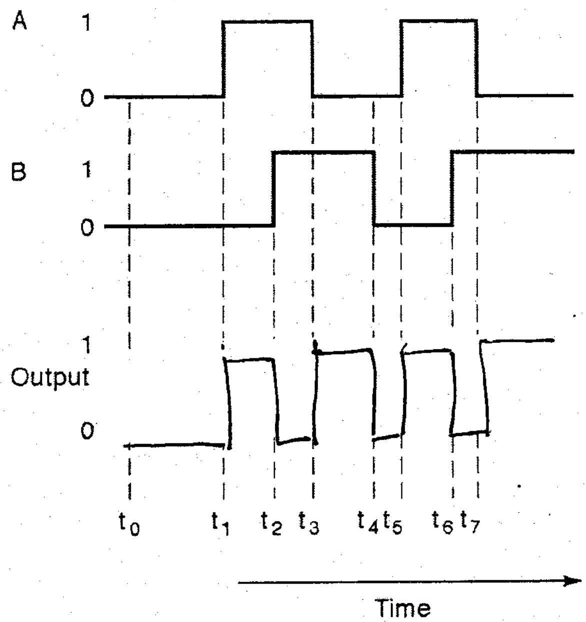
2. Write down the truth table for the expression $z = A\overline{B}\,\overline{C} + A\overline{B}C + ABC$. Further simplify this expression using K-map. (3 Marks)

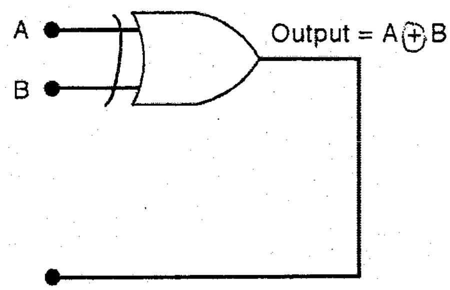
A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

AB \ C	0	1
00		
01		
11		1
10	1	1

$$Z = A\overline{B} + AC$$

3. a) Determine the output waveform for an XOR gate given its input waveform as shown below. (2 Marks)





b) For 8 bit transmission including the Parity bit- (2 Mark)

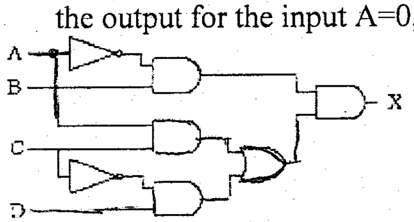
i) Determine the parity bit for even parity transmission for the data 1011010.

0 1011010

ii) Determine the parity bit for odd parity transmission for the data 1111011.

1 1111011

4. Write the expression for the output for the logic circuit shown below. Determine the output for the input A=0, B=1, C=1 and D=0 (3 Marks)



$= \bar{A}B(AC + \bar{C}D)$

Figure 3-9

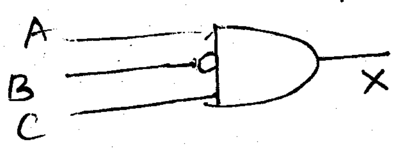
$X = 1.1(0.1 + 0.0)$

$= 1.0$

$= 0$

6. A) Design a logic circuit that will allow a signal A to pass through to the output only when control input B is LOW while control input C is HIGH; otherwise the output is LOW. (2 marks)

$X = A\bar{B}C$



B) Draw the standard logic symbol of AND gate. What is the active state for the output and inputs. (1 Marks)

output is HIGH WHEN INPUTS ARE HIGH

ACTIVE STATES = 1 or HIGH FOR INPUTS & OUTPUT

7. For the BCD to 5421 code converter, i) determine the SOP expression for the output bit that has weight 4. ii) determine the POS expression for the output that has weight 2. (4 Marks)

ABCD	WXYZ
8 4 2 1	5 4 2 1
0 0 0 0	0 0 0 0
0 0 0 1	0 0 0 1
0 0 1 0	0 0 1 0
0 0 1 1	0 0 1 1
0 1 0 0	0 1 0 0
0 1 0 1	1 0 0 0
0 1 1 0	1 0 0 1
0 1 1 1	1 0 1 0
1 0 0 0	1 0 1 1
1 0 0 1	1 1 0 0

Weight 4 = X

$X = \Sigma(4, 9) + \Sigma(10, 11, 12, 13, 14, 15)$ Don't Cares

$X = \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}D$ + Don't cares

AB \ CD	00	01	11	10
00				
01	1			
11	X	X	X	X
10		1	X	X

$X = AD + B\bar{C}\bar{D}$ ← SOP form

Weight 2 = Y

$Y = \Pi(0, 1, 4, 5, 6, 9)$

$= (A+B+C+D)(A+B+C+\bar{D})(A+\bar{B}+C+D)\cdot$
$(A+\bar{B}+C+\bar{D})(A+\bar{B}+\bar{C}+D)(\bar{A}+B+C+\bar{D})$

$\bar{Y}$ or K-Map with Don't cares

AB \ CD	00	01	11	10
00	0	0		0
01	0	0		0
11	X	X	X	X
10		0	X	X

$\bar{Y} = \bar{A}\bar{C} + \bar{C}D + B\bar{D}$

$\bar{\bar{Y}} = Y = \overline{\bar{A}\bar{C} + \bar{C}D + B\bar{D}} = \overline{\bar{A}\bar{C}} \cdot \overline{\bar{C}D} \cdot \overline{B\bar{D}}$

$= (A+C)\cdot(C+\bar{D})\cdot(\bar{B}+D)$

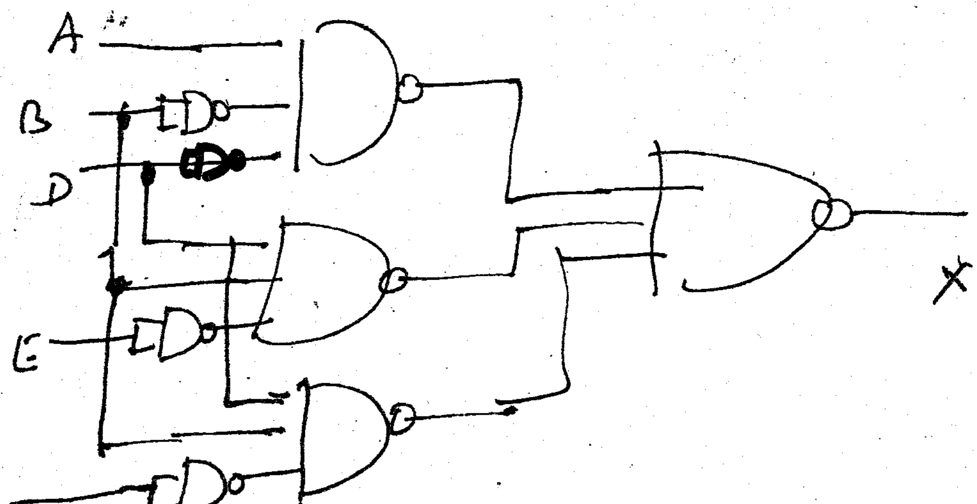
Another possible solution is

$Y = (A+C)(\bar{B}+D)(\bar{A}+\bar{D})$

8. Implement the following expression using NAND GATES ONLY
$x = A\bar{B}\bar{D} + BD\bar{E} + B\bar{C}D$ (3 Marks)

$$x = \overline{\overline{A\bar{B}\bar{D} + BD\bar{E} + B\bar{C}D}}$$

$$= \overline{\overline{A\bar{B}\bar{D}} \cdot \overline{BD\bar{E}} \cdot \overline{B\bar{C}D}}$$



9. Implement the following expression using NOR GATES ONLY
$x = (A + \bar{B})(\bar{A} + C)(B + \bar{D})$ (3 Marks)

$$x = \overline{\overline{(A+\bar{B})(\bar{A}+C)(B+\bar{D})}}$$

$$= \overline{\overline{(A+\bar{B})} + \overline{(\bar{A}+C)} + \overline{(B+\bar{D})}}$$